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Preparing Effective Patent Proposal

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What is an invention disclosure form?

- Document used to capture detailed information about a new invention or discovery.
- Document in which an innovator records their discovery, including any supporting data and results.
- It must provide sufficient information
 - to permit the patent controller to assess whether the invention in question has merit and/or commercialization potential
 - to permit patent attorney to prepare a detailed patent application

Title of the Invention:

- The title needs to be precise, specific, and no more than **15 words**.
- The title should describe what the Invention does, but not how it is made or how it works.
- Clearly indicate whether the invention is a **product, process, apparatus, or composition of matter**.
 - *Example:* "Method for manufacturing..." or "Apparatus for detecting..."
- **Example:** Solar-Powered Backpack with Integrated Water Purification System
- Avoid words like "**new,**" "**improved,**" "**revolutionary,**" or "**unique**".

Technical Field of the Invention:

- Define the specific technology domain, industry, or area of application to which the invention pertains.
- Begin with a general field and progressively define the specific area.
 - **Example:** "The present invention relates to the field of renewable energy, more specifically to generate power and purify water simultaneously".
- Clearly identify the technical domain (e.g., IoT, artificial intelligence, software, mechanical engineering, biotechnology) or industry sector (e.g., automotive, medical devices).

Background of the Invention:

- Address the gap and Novelty.
- Briefly describe the problem your invention addresses.
- Explain existing solutions and their limitations.
- Highlight the novelty and advantages of your invention.
- **Example:** Existing solar-powered backpacks lack integrated water purification capabilities, and portable water purifiers are often bulky and inconvenient. This invention addresses these limitations by combining solar panels with a UV water purification system within a backpack, enabling off-grid adventurers to generate power and purify water simultaneously.

Prior Art:

- Writing the prior art section requires thorough identification of existing technologies, patents, and publications that predate your invention.
- List patent numbers, technical articles, journal and/or conference publications that are relevant (similar) to your invention.
- Detail the differences between your invention and the closest prior art. Highlight why your solution is better, such as higher efficiency, lower cost, or a novel method of operation.

Prior Art Example:

- Patent ID: IN 202041055532
- Title: IoT Based LPG Gas Leakage Detection and Monitoring System
- This invention reveals an IoT-connected gas leakage detector with gas sensors of MQ-series and linked to a microcontroller to monitor real-time gas concentration. When a leakage is identified beyond a certain threshold, the system activates a local buzzer and sends mobile notifications either through Wi-Fi or GSM. Nevertheless, this system lacks overfill detection by weight, cylinder expiry validation, AI-based lifecycle prediction, digital identity, autonomy shut-off of solenoid, and distribution optimization. It works in a reactive way and does not have preventative analytics and is incapable of anticipating and averting dangerous states before they emerge.

Prior Art Example:

- Patent ID: US 10,690,327 B2
- Title: Gas Cylinder Monitoring and Control System
- This US patent reveals a gas cylinder monitoring and control system that includes pressure sensors, gas detectors and remote notification. The system allows the remote monitoring and alerts when the abnormal pressure condition is detected. Nevertheless, it is not mentioning the overfill detection based on weight, AI-based refill prediction, digital identity assignment and tracking of cylinders, expiry or re-test date verification before delivery, multi-stage leakage identification with emergency autonomous shut-off, and intelligent distribution and vehicle routing management. Unified monitoring of safety and logistics that are central to the system is not provided.

Objective(s) of Invention:

- What are you hoping to achieve with your invention?
- What specific goals does it address?
- Example:
 1. To provide a self-sufficient backpack for outdoor activities.
 2. To offer clean drinking water during travel and in remote locations.
 3. To promote sustainable energy use by harnessing solar power.

Synopsis:

- Brief summary of your invention.
- If it is a device/system what the device/system is?
- Components required for construction of the device/system.
- How the components are assembled for construction of the device/system.
- **Example:** Solar panels on the backpack convert sunlight into electricity, stored in a battery. This powers the UV water purification system integrated within the backpack's water reservoir. Users simply fill the reservoir with water from any source, activate the UV system, and clean drinking water is dispensed through a dedicated tap.

Brief Description of Drawings (Example 1):

- If any figures are given to describe the invention, only numerals must be used on it and what numerals denote must be described here. Don't keep figures in this section. Keep the figures in last Drawings section.
- Figure 1 illustrates basic diagram of proposed invention. Whereas,
 - 100 denotes top surface,
 - 101 denotes ratchet hinge mechanism,
 - 102 denotes upper hollow shaft,
 - 103 denotes vertical locking mechanism,
 - 104 denotes lower hollow shaft,
 - 105 denotes base part,
 - 106 denotes legs of base

Figure 1 Drawing (on last page of the proposal)

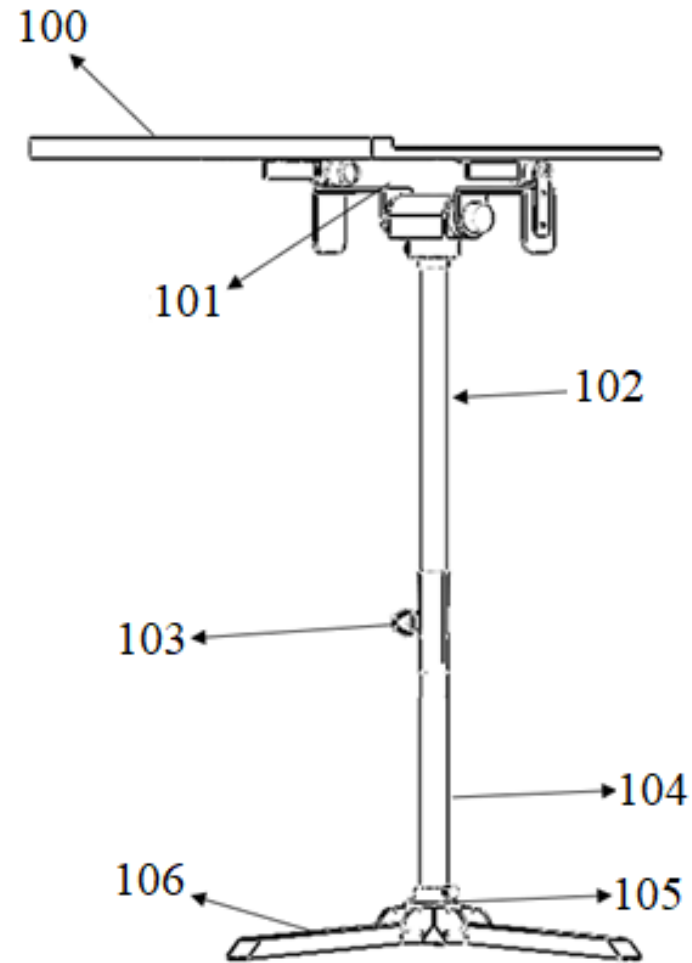


Figure 1

Brief Description of Drawings (Example 2):

- Figure 2 illustrates system layout of proposed invention. Whereas,
100 denotes smartphone,
101denotes server,
102 denotes solenoid valve
- Figure 2A illustrates circuit of proposed invention. Whereas,
- 103 denotes node MCU,
- 104 denotes relay module

Figure 2 and 2A Drawings (on last page of the proposal)

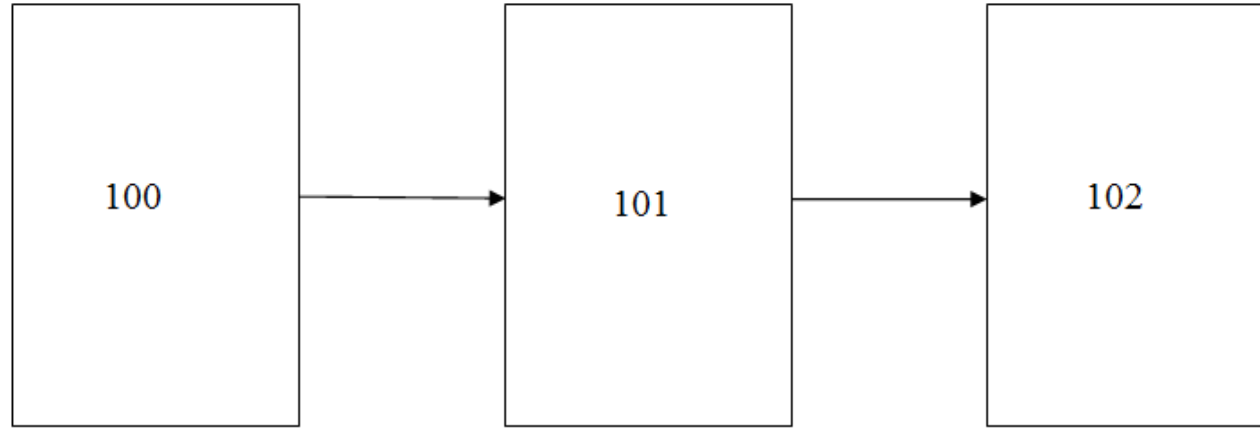


Figure 2

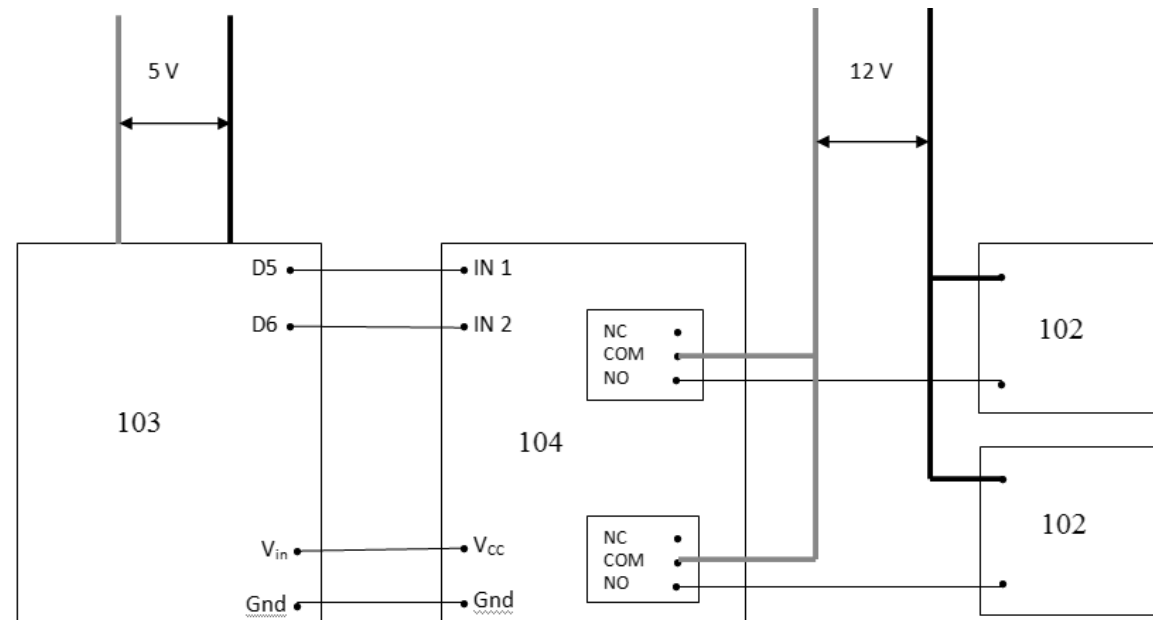


Figure 2A

Detailed Description of the Invention:

- This section should be comprehensive, covering the composition, structure, and functional working of the invention.
- List all physical parts (for devices) or steps (for processes/methods).
- Explain how parts are connected or how process steps are sequenced.
- Describe how the invention operates to achieve its intended result.
- Use ordinary technical terms, but provide definitions for any unusual terms.

Best Method of Performance of the Invention:

- Disclose your preferred way of carrying out the invention at the time of filing.
- Preferably in one to two paragraphs mention the process in which invention is performed by giving example. (Actual practical working steps of your invention)
- Describe any tests or experiments conducted to validate the effectiveness of your invention.
- Share the results and their significance.

Claims:

- Mention what aspect(s) of the invention need(s) protection.
- Specify the unique features and functionalities of your invention that you wish to protect through patent. This could include specific components, the combination of elements, or the overall operating principle.
- Example: (Claims for invention shown in Figure 2 and 2A)
 1. An IOT based valve control system, comprising:
 - a solenoid valve [102] adapted to control the flow rate of water flowing through piping system;

Claims:

a node MCU [103] connected with the solenoid valve [102], to control the operation of said solenoid valve [102];

a moisture sensor adapted to sense the moisture content in the soil and help operator to make decision about water supply to field; and

a relay module [104] is connected in between to separate low voltage side and high voltage side in the circuit.

2. The IOT based valve control system as claimed in claim 1, wherein the said relay module [104] is connected to the solenoid valves [102] in normally open (NO) mode.
3. The IOT based valve control system as claimed in claim 1, said solenoid valve [102] gets actuated from the relay [104] and is turned ON or OFF according to the signal sent by the operator [100].

Inventive Step of your Invention:

- Highlight what are the innovative features of your invention which makes your invention different than present technology.
- For patentability, your invention must achieve either technical advantage over existing technology or it must reduce the cost of existing technology.
- Mention HOW this advantage is achieved by your invention.

Industrial Application(s):

- In which field or purpose your invention is useful?
- Define the precise industry sector (e.g., automotive, pharmaceutical, software, manufacturing) and the specific area where the invention will be used.
- **On the top:** Identify Potential Users and Markets: List specific companies, industries, or customers who would benefit from or license the technology

Abstract:

- Just write basic idea about your invention in one paragraph.
- **Example:**

The present invention relates an IOT based valve control system. The object of the proposed invention is to reduce physical efforts of farmers. The system consists of two subsystems; actuating system is physical subsystem consisting of valve actuating mechanism, valves and moisture measurement system. It is installing in the field itself and virtual control system is the virtual subsystem which consists of mobile application or website. It is installing in operator's mobile phone. Wi-Fi (Wireless Fidelity) or GSM (Global system for mobile) based signal system connects actuating and virtual control system. Following invention is described in detail with the help of Figure 1A which illustrates circuit diagram of proposed invention.

Drawing(s):

- **Types of Drawings Needed in Black and White**
 - **Block Diagrams:** Ideal for software, systems, or process inventions to show the interaction between different components (e.g., Input → Processor → Output)
 - **Flowcharts:** Essential for process, method, or software inventions. Map out each step from start to finish, including decision points.
 - **Structural Sketches/Line Drawings:** For hardware, machines, or devices. Include top, side, front, and perspective views.
 - **Exploded Views:** Show how individual parts fit together to form the complete assembly.
 - **Cross-Sectional Views:** Used to explain internal structures or hidden mechanisms.

